4.1 Classifying Triangles & Angle Sum

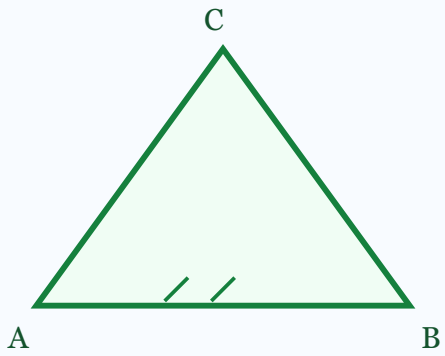
Triangles are classified two independent ways. **By sides:** scalene (no two sides congruent), isosceles (at least two sides congruent), equilateral (all three sides congruent). **By angles:** acute (all three angles <90^\circ), right (exactly one $90°$ angle), obtuse (exactly one angle >90^\circ), equiangular (all three angles congruent, which forces equilateral). The **Triangle Angle-Sum Theorem** guarantees the interior angles of every triangle total $180°$; its corollary, the **Exterior Angle Theorem**, says an exterior angle equals the sum of the two non-adjacent (remote) interior angles. Because the angle sum is fixed at $180°$, a triangle can never have two right angles or two obtuse angles, and the acute angles of a right triangle must be complementary. This worksheet blends definitions, algebra, coordinate geometry, and short proofs.

Examples

Example 1: Classify by sides

A triangle has sides 8, 8, and 13. Classify it by its sides.

Solution



Tick marks show the two congruent sides.

Compare all three side lengths: 8, 8, 13.

Two of the three sides are congruent ($8 = 8$); the third is different.

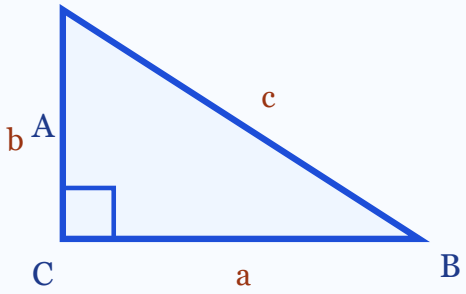
This matches the definition of isosceles (not equilateral, since not all three match; not scalene, since two match).

Answer: Isosceles.

Example 2: Classify by angles

A triangle has angles $38°$, $52°$, and $90°$. Classify it by its angles, and check the angle sum.

Solution



One angle is exactly $90°$, so this is a right triangle.

Check the sum: $38 + 52 + 90 = 180$ ✓ (valid triangle).

One angle equals exactly $90°$, so by definition this is a **right** triangle.

Note the two acute angles are complementary: $38 + 52 = 90$, as they must be in any right triangle.

Answer: Right triangle.

Example 3: Angles in a given ratio

The three angles of a triangle are in the ratio $2 : 3 : 4$. Find each angle, then classify the triangle by angles.

Solution

Let the angles be $2x$, $3x$, and $4x$.

Apply the Angle-Sum Theorem: $2x + 3x + 4x = 180$

$9x = 180 \Rightarrow x = 20$.

Angles: $2(20) = 40°$, $3(20) = 60°$, $4(20) = 80°$.

Check: $40 + 60 + 80 = 180$ ✓. All three angles are less than $90°$.

Answer: $40°$, $60°$, $80°$ — acute triangle.

Example 4: Algebraic angle sum

In $\triangle ABC$, $\angle A = 3x + 10$, $\angle B = 2x - 4$, and $\angle C = 104°$. Find x and each angle, then classify by angles.

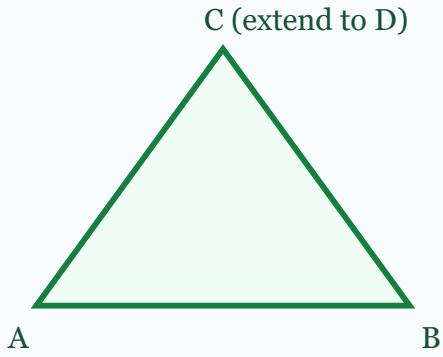
Solution

Identify the relationship: the three angles must sum to $180°$.
Set up the equation: $(3x + 10) + (2x - 4) + 104 = 180$
Combine like terms: $5x + 110 = 180$
Solve: $5x = 70 \Rightarrow x = 14$.
Substitute back: $\angle A = 3(14) + 10 = 52°$, $\angle B = 2(14) - 4 = 24°$, $\angle C = 104°$.
Check: $52 + 24 + 104 = 180$ ✓. Since \angle C>90^{\circ}, the triangle is obtuse.
Answer: $x = 14$; angles $52°, 24°, 104°$ — obtuse triangle.

Example 5: Prove the Exterior Angle Theorem

Prove that an exterior angle of a triangle equals the sum of the two remote interior angles.

Solution



Extend a side of the triangle to form exterior $\angle ACD$.

Let the triangle's interior angles be $\angle A$, $\angle B$, $\angle C$, and extend side $\overline{BC}$ through C to point D, forming exterior angle $\angle ACD$.
By the Angle-Sum Theorem: $\angle A + \angle B + \angle C = 180°$, so $\angle A + \angle B = 180° - \angle C$.
Because $\angle ACD$ and $\angle C$ form a linear pair (they lie on straight line BD): $\angle ACD + \angle C = 180°$, so $\angle ACD = 180° - \angle C$.
Both expressions equal $180° - \angle C$, so $\angle ACD = \angle A + \angle B$. ■
Conclusion: the exterior angle equals the sum of the two remote (non-adjacent) interior angles.

Example 6: Apply the Exterior Angle Theorem

An exterior angle of a triangle measures $130°$. One remote interior angle is $55°$. Find the other remote interior angle and the interior angle adjacent to the exterior angle.

Solution

By the Exterior Angle Theorem, the exterior angle equals the sum of the two remote interior angles:

$130 = 55 + (\text{other remote})$

Other remote angle $= 130 - 55 = 75°$.

The adjacent interior angle is supplementary to the exterior angle (linear pair): $180 - 130 = 50°$.

Check all three interior angles: $55 + 75 + 50 = 180$ ✓.

Answer: Other remote angle $= 75°$; adjacent interior angle $= 50°$.

Example 7: Exterior angle with two unknowns

An exterior angle of a triangle measures $4x$. The two remote interior angles measure $2x + 10$ and $x + 20$. Find x and the exterior angle.

Solution

Apply the Exterior Angle Theorem: exterior = sum of remotes.

$4x = (2x + 10) + (x + 20)$

$4x = 3x + 30$

$x = 30$.

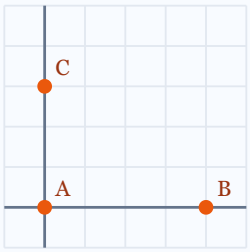
Exterior angle $= 4(30) = 120°$. Check remotes: $2(30) + 10 = 70$ and $30 + 20 = 50$; $70 + 50 = 120$ ✓.

Answer: $x = 30$; exterior angle $= 120°$.

Example 8: Classify a coordinate triangle

Triangle has vertices $A(0, 0)$, $B(4, 0)$, $C(0, 3)$. Find all three side lengths and classify the triangle by both sides and angles.

Solution



Right angle appears at A, where the legs meet the axes.

Use the distance formula for each side:

$AB = \sqrt{(4-0)^2 + (0-0)^2} = \sqrt{16} = 4$

$AC = \sqrt{(0-0)^2 + (3-0)^2} = \sqrt{9} = 3$

$BC = \sqrt{(0-4)^2 + (3-0)^2} = \sqrt{16+9} = \sqrt{25} = 5.$

By sides: $3, 4, 5$ are all different $\Rightarrow$ scalene.

By angles: test the converse of the Pythagorean Theorem on the longest side: $3^2 + 4^2 = 9 + 16 = 25 = 5^2$ ✓, so the angle at A is right.

Answer: Scalene right triangle.

Example 9: Why two obtuse angles is impossible

Explain why a triangle cannot have two obtuse angles.

Solution

Suppose, for contradiction, a triangle had two obtuse angles, each greater than $90°$.
Their sum alone would already exceed $90° + 90° = 180°$.
But the full angle sum of any triangle is exactly $180°$, and angle measures must be positive.
Adding a third positive angle would push the total above $180°$, contradicting the Angle-Sum Theorem.
Conclusion: a triangle can have at most one obtuse angle (and, by the same reasoning, at most one right angle).

Example 10: Isosceles triangle with an exterior angle

In isosceles $\triangle ABC$ with $AB = AC$, the exterior angle at base vertex B measures $130°$. Find $\angle B$, $\angle C$, and vertex angle $\angle A$.

Solution

Identify: $\angle B$ and its exterior angle form a linear pair, so they are supplementary.
$\angle B = 180 - 130 = 50°$.
Since $AB = AC$, the base angles $\angle B$ and $\angle C$ are congruent (isosceles base angles): $\angle C = 50°$.
Use the Angle-Sum Theorem for the vertex angle: $\angle A = 180 - 50 - 50 = 80°$.
Check: $50 + 50 + 80 = 180$ ✓; all angles acute, so the triangle is acute isosceles.
Answer: $\angle B = 50°$, $\angle C = 50°$, $\angle A = 80°$.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Classify by sides

A triangle has sides 9, 9, 9. Classify it by sides.

Answer: **Verification:**

Problem 2: Classify by sides

A triangle has sides 6, 9, 11. Classify it by sides.

Answer: **Verification:**

Problem 3: Classify by angles

A triangle has angles 100°, 40°, 40°. Classify by angles.

Answer: **Verification:**

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Problem 4: Classify by angles

A triangle has angles $90°$, $45°$, $45°$. Classify by angles.

Answer: **Verification:**

Problem 5: Angle sum

A triangle has angles $65°$, $75°$, and x. Find x.

Answer: **Verification:**

Problem 6: Ratio of angles

The angles of a triangle are in ratio $1:2:3$. Find each angle and classify by angles.

Answer: **Verification:**

Problem 7: Algebra

In $\triangle ABC$, $\angle A = 2x + 5$, $\angle B = 3x - 1$, $\angle C = 91°$. Find x, each angle, and classify by angles.

Answer: **Verification:**

Problem 8: Exterior angle

The two remote interior angles are $48°$ and $62°$. Find the exterior angle.

Answer: **Verification:**

Problem 9: Exterior angle

An exterior angle is $125°$; one remote interior angle is $70°$. Find the other remote angle.

Answer: **Verification:**

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Problem 10: Exterior algebra

Exterior angle = $4x$; remote angles are $2x + 10$ and $x + 20$. Find x and the exterior angle.

Answer: **Verification:**

Problem 11: Coordinate triangle

Vertices $A(0, 0)$, $B(6, 0)$, $C(0, 8)$. Find all three side lengths and classify by sides and angles.

Answer: **Verification:**

Problem 12: Justify

Explain why a triangle cannot have two right angles.

Answer: **Verification:**

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Problem 13: Challenge — isosceles + exterior

In isosceles $\triangle ABC$ with $AB = AC$, the exterior angle at B measures $150°$. Find $\angle B$, $\angle C$, and $\angle A$, then classify by angles.

Answer: **Verification:**

Problem 14: Challenge — coordinate classification

Vertices $A(1, 1)$, $B(5, 1)$, $C(1, 5)$. Find all side lengths (exact form) and fully classify the triangle by sides and angles.

Answer: **Verification:**

Problem 15: Challenge — exterior word problem

In $\triangle ABC$, $\angle A = \angle B$ (isosceles), and the exterior angle at C measures $100°$. Find $\angle C$, then $\angle A$ and $\angle B$, and classify by angles.

Answer: **Verification:**

✓ Answer Key

1. Equilateral
2. Scalene
3. Obtuse
4. Right
5. $x = 40°$
6. $30°, 60°, 90°$ — right triangle
7. $x = 17$; angles $39°, 50°, 91°$ — obtuse triangle
8. $110°$
9. $55°$
10. $x = 30$; exterior angle $= 120°$
11. $6, 8, 10$; right scalene triangle
12. Two $90°$ angles already total $180°$, leaving no positive measure for the third angle — contradicting the Angle-Sum Theorem
13. $\angle B = \angle C = 30°$, $\angle A = 120°$ — obtuse isosceles
14. $AB = AC = 4$, $BC = 4\sqrt{2}$; isosceles right triangle (right angle at A)
15. $\angle C = 80°$, $\angle A = \angle B = 50°$ — acute isosceles

Great work — keep practising!